

- 6 (a) In mammals, the blood glucose concentration must be maintained within narrow limits so that the body cells can function efficiently.

Name the mechanism by which the blood glucose concentration is maintained within narrow limits.

..... [1]

- (b) Glucagon is released by the alpha (α) cells of the pancreas when the blood glucose concentration decreases below the set point.

Fig. 6.1 outlines the response of liver cells to glucagon.

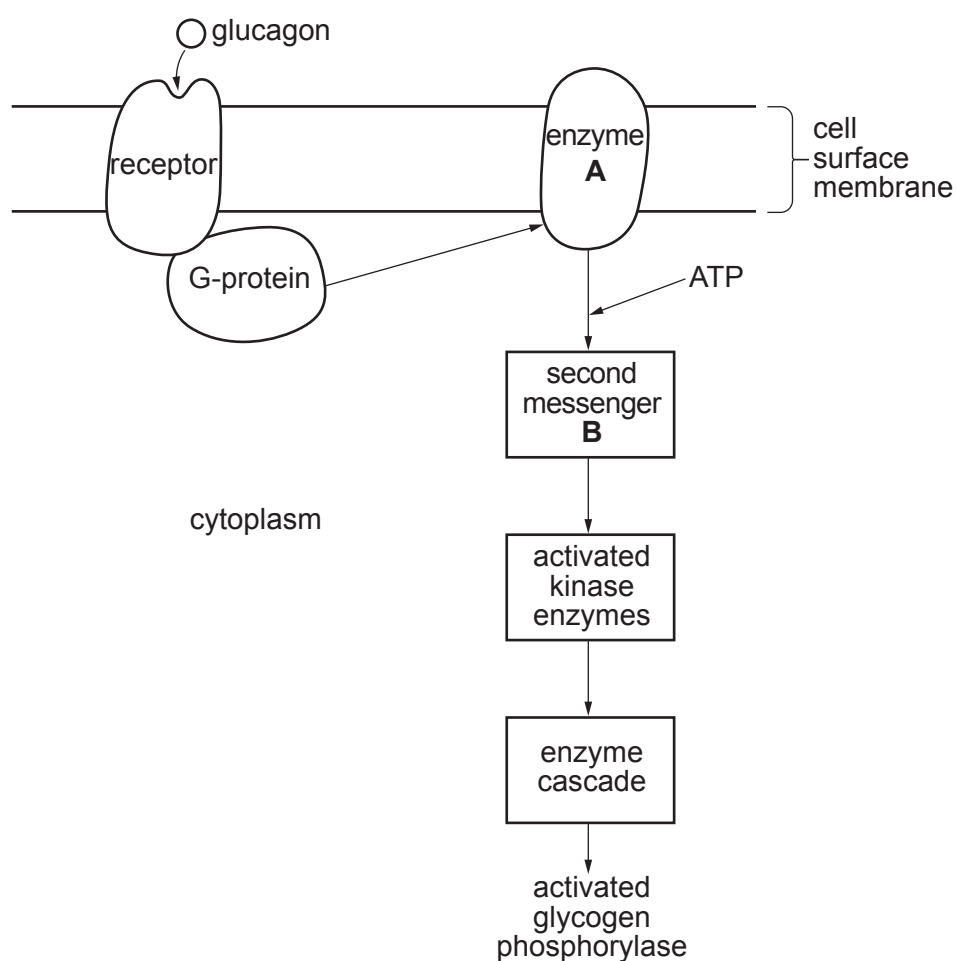


Fig. 6.1

- (i) State how glucagon reaches the liver cells.

..... [1]

- (ii) With reference to Fig. 6.1, name enzyme **A** and second messenger **B**.

A

B

[2]

(iii) State the role of the enzyme cascade.

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..... [1]

(iv) State the function of the final enzyme in the pathway, glycogen phosphorylase.

..... [1]

(c) A biosensor is used to measure blood glucose concentration to check that it is within the normal range.

Describe how a glucose biosensor works.

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..... [4]

[Total: 10]